

№ 5.54

$$f = aT \left(\frac{L}{L_0} - \left(\frac{L_0}{L} \right)^2 \right) \quad \begin{array}{l} \text{— растягивающая сила.} \\ L \text{ — длина} \\ L_0 \text{ — нач. длина тн.} \end{array}$$

$$a = 0,013 \text{ Н/К}$$



$$1-2: T = \text{const} = 300 \text{ K} \quad C_L = 1,2 \frac{\text{Дж}}{\text{К}}$$

$$2-3: L = \text{const} \text{ картел}$$

$$3-1: \delta Q = 0, \text{ сжатие}$$

$$Q_{12} = ? \quad Q_{23} = ?$$

$$1) dS(T, V) = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV = \frac{C_V}{T} dT + \left(\frac{\partial P}{\partial T} \right)_V dV$$

$$= \frac{C_V}{T} dT - \left(\frac{\partial f}{\partial T} \right)_L \frac{1}{\sigma} dV \Rightarrow$$

$$\Rightarrow dS = \frac{C_V}{T} dT - \left(\frac{\partial f}{\partial T} \right)_L dL$$

$$1 \rightarrow 2: \left(\frac{\partial f}{\partial T} \right)_L = a \left(\frac{L}{L_0} - \frac{L_0^2}{L^2} \right) \rightarrow \Delta S_{12} = - \frac{a}{L_0} \int_{1-2} L \left(1 + 0 \right) dL \int_{1-2} \frac{dT}{T}$$

$$= a \left(\frac{L_0^2 - L^2}{2L_0} + L_0^2 \left(\frac{1}{L_0} - \frac{1}{L} \right) \right) = \dots = -0,013 \frac{\text{Дж}}{\text{К}} < 0$$

$$Q_{12} = T \Delta S_{12} = -3,9 \text{ Дж}$$

$$2 \rightarrow 3: dS = \frac{C_L}{T} dT$$

$$\Delta S_{2-3} = -\Delta S_{12} = \frac{C_L}{T_2} \ln \frac{T_3}{T_2}$$

$$T_3 = T_2 \exp \left(\frac{\Delta S_{23}}{C_L} \right) = T_2 e^{\frac{-\Delta S_{12}}{C_L}} = \dots = 303 \text{ K}$$

Т.4.1.

$$C_V = aVT^3 \quad (C_P - C_V(V, T) = ?)$$

$$\kappa = \frac{1}{\beta_T}, \quad \beta_T = \frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$$

$$C_P - C_V = T \left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_P = T \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P = T \left(\frac{\partial S}{\partial V} \right)_T^2$$

$$\frac{\partial P}{\partial T} \frac{\partial T}{\partial V} \frac{\partial V}{\partial P} = -1 \Rightarrow \left(\frac{\partial V}{\partial T} \right)_P = - \left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial P} \right)_T = - \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{\partial V}{\partial P} \right)_T$$

$$\text{Тогда: } C_P - C_V = -T \left(\frac{\partial S}{\partial V} \right)_T^2 \left(\frac{\partial V}{\partial P} \right)_T = - \frac{TV}{\kappa} \left(\frac{\partial S}{\partial V} \right)_T^2 = S = \int \frac{C_V dT}{T} = \int aVT^3 dT = aV \frac{T^3}{3}$$

$$= \frac{TV}{\kappa} \cdot \frac{a^2 T^6}{9} = \frac{a^2 VT^7}{9\kappa} \quad \frac{\partial}{\partial V} \left(aV \frac{T^3}{3} \right) = \frac{aT^3}{3}$$

№ 5.63

$$\frac{\Delta V}{V} = 0,01$$

$$\frac{\Delta T}{T} = 0,028$$

$$\beta_T = ?$$

$$\alpha = 5,7 \cdot 10^{-5} \text{ К}^{-1}$$

$$C_V = 0,23 \frac{\text{Дж}}{\text{г} \cdot \text{К}}$$

$$\alpha = \left(\frac{\partial P}{\partial T} \right)_V$$

$$\beta_T = - \frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$$

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

$$\left(\frac{\partial P}{\partial T} \right)_V = \left(\frac{\partial S}{\partial V} \right)_T = - \left(\frac{\partial S}{\partial T} \right)_V \left(\frac{\partial T}{\partial V} \right)_S = \frac{C_V}{T} \left(\frac{\partial T}{\partial V} \right)_S$$

$$\frac{\partial S}{\partial V} \frac{\partial V}{\partial T} \frac{\partial T}{\partial S} = -1$$

$$\alpha = \beta_T \left(\frac{\partial P}{\partial T} \right)_V = - \beta_T \left(\frac{\partial T}{\partial V} \right)_S \frac{C_V}{T} \cdot \frac{V}{V} =$$

$$= - \beta_T C_V \left(\frac{\Delta T}{T} \right)_S \left(\frac{\Delta V}{\Delta V} \right) \cdot \frac{1}{V} = - \beta_T C_V \left(\frac{\Delta T}{T} \right)_S \frac{P}{\rho}$$

$$\beta_T = - \frac{\alpha \cdot m}{C_V \rho} \cdot \frac{\Delta V}{\Delta T} = \dots = 8,4 \cdot 10^{12} \text{ Па}^{-1}$$

№ 5.32

1 Мол

$$\psi = - \frac{RT}{2} \ln(AT^3 V^2)$$

$$C_P = ?$$

$$d\psi = -SdT - PdV$$

$$\left(\frac{\partial \psi}{\partial V} \right)_T = -P; \quad \left(\frac{\partial \psi}{\partial T} \right)_V = -S$$

$$-P = \left(\frac{\partial \psi}{\partial V} \right)_T = \frac{RT}{2} \frac{1}{AT^3 V^2} \cdot 2AT^3 V^2 = - \frac{RT}{V}$$

$$PV = RT$$

$$\sqrt{12.7}$$

$$\delta Q = 0$$

$$S(T, F) = \left(\frac{\partial S}{\partial T} \right)_F dT + \left(\frac{\partial S}{\partial F} \right)_T dF = 0$$

$$\frac{dT}{dF} = - \frac{\left(\frac{\partial S}{\partial F} \right)_T}{\left(\frac{\partial S}{\partial T} \right)_F} = \frac{\frac{d\sigma}{dT}}{C_F/T} = \frac{T}{C_F} \frac{d\sigma}{dT}$$

$$= - \frac{q}{C_F}$$

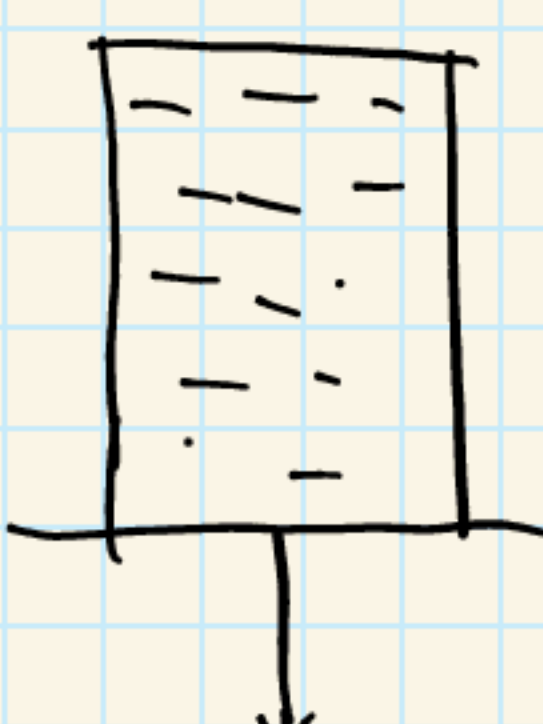
$$\sqrt{12.8}$$

$$h = 10^{-3} \text{ мм}$$

$$T = 300 \text{ K}$$

$$F = 2F_0$$

$$\frac{d\sigma}{dT} = -0,15 \frac{\text{Дж}}{\text{г} \cdot \text{К}}$$



$$\left(\frac{dT}{dF} \right)_S = \frac{T}{C_F} \frac{d\sigma}{dT} =$$

$$= - \frac{q}{C_F}$$

$$C_F = C_{yg} = \left(\sqrt{\pi} \frac{h}{2} \right) \ln 1 = 4,2 \cdot 10^{-7} \frac{\text{Дж}}{\text{г} \cdot \text{К}}$$

$$= C_{yg} \cdot gV = C_{yg} \rho h \frac{F_0}{2}$$

$$\Delta T_s = \frac{2T}{C_{yg} \rho h F_0} (2F_0 - F_0) = \dots = -0,02 \text{ К}$$